

Topic 1 The Boltzmann Distribution

Goals:

understand relations between

$$\begin{bmatrix} \text{energy } E \\ \text{temperature } T \\ \text{entropy } S \\ \text{probability } p \end{bmatrix}$$

and derive the Boltzmann distribution:

$$p(\vec{s}) = \frac{1}{Z} e^{-\frac{E(\vec{s})}{kT}}$$
$$Z = \int d\vec{s} e^{-E(\vec{s})/kT}$$

$\vec{s}$ = microstate of physical system

$E(\vec{s})$ = energy of microstate

T = temperature

k = Boltzmann's constant

kT = units of energy

Step 1: understand statistical mechanics of closed systems

- a closed system is isolated from rest of universe
- this means its energy is conserved
- but there may be many microstates $\vec{s}$ with a given energy E



$$\Omega(E) dE = \# \text{ of microstates } \vec{s} \text{ w/ energy } E(\vec{s}) \text{ between } E \text{ and } E + dE$$

- Usually, in systems of N particles or spins, $\Omega(E)$ is exponential in N
- this motivates taking a logarithm to obtain entropy:

$$S(E) = \ln \Omega(E)$$

Definition of entropy of a closed system
with energy E

$$\Omega(E) = e^{S(E)}$$

- if there are many microstates $\vec{s}$ with energy $E(\vec{s}) = E_0$

which one will you find the system in if you measure it in any given time

Assumption 1 Microcanonical Ensemble for a closed system

you will find the system in a random state $\vec{s}$ drawn from a uniform distribution $p(\vec{s})$ over all states $\vec{s}$ with energy E_0

$$\text{ie } p(\vec{s}) = \frac{1}{\Omega(E_0)} \delta[E(\vec{s}) - E_0]$$

- intuition
- as the system evolves the state $\vec{s}(t)$ will change with time t Starting from some random state
 - you may want to calculate average quantities of functions of $\vec{s}$

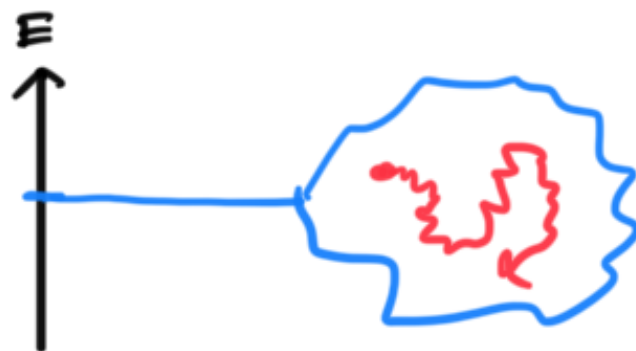
ie
for large
enough T

$$\frac{1}{T} \int dt F(\vec{s}(t)) dt \longleftrightarrow \int d\vec{s} F(\vec{s}) p(\vec{s})$$

time average

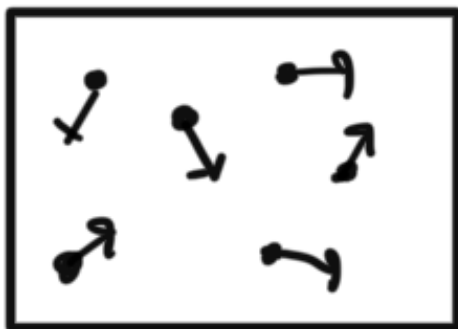
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ensemble average



$\vec{s}(t)$ starts in a random location
or evolves chaotically or randomly
over energy surface

Example: Ideal gas



the microstate is the positions $\vec{x}_i$ for all $i=1 \dots N$
 momenta $\vec{p}_i$ particles

$\vec{s} = [\vec{p}_1, \dots, \vec{p}_N, \vec{x}_1, \dots, \vec{x}_N]$ a $6N$ dimensional vector

- recall momentum of a particle with velocity $\vec{v}$: $\vec{p} = m\vec{v}$
 and mass m
 kinetic energy $E = \frac{1}{2} m \vec{v}^2 = \frac{\vec{p} \cdot \vec{p}}{2m}$

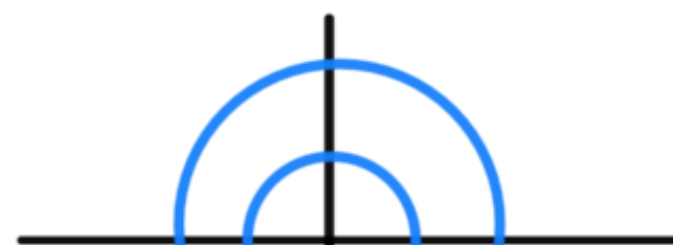
- so energy of ideal gas:

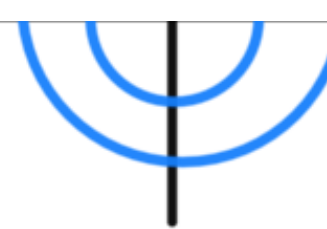
$$E(\vec{s}) = \sum_{i=1}^N \frac{\vec{p}_i \cdot \vec{p}_i}{2m}$$

- set of microstates with energy E is a sphere of radius $\sqrt{2mE}$
 in momentum space $\vec{p}$ and position just has to be in box:

$[\vec{p}_1, \dots, \vec{p}_N]$

$$\sum_{i=1}^N \vec{p}_i \cdot \vec{p}_i = 2mE$$





$$\Omega(E) = \int d\vec{p}_1 \dots d\vec{p}_N d\vec{x}_1 \dots d\vec{x}_N \delta \left[\sum_{i=1}^N \vec{p}_i \cdot \vec{p}_i - 2mE \right]$$

$$= \underbrace{\int d\vec{p}_1 \dots d\vec{p}_N \delta \left[\sum_{i=1}^N \vec{p}_i \cdot \vec{p}_i - 2mE \right]}_{\text{volume of sphere}} \underbrace{\int_{\text{Box}} d\vec{x}_1 \dots d\vec{x}_N}_{V^N \quad V = \text{volume of box}}$$

recall a sphere of radius r in dim n

is $\frac{(2\pi)^{n/2}}{\Gamma(n/2)} r^{n-1}$ plug in $r = \sqrt{2mE}$
 $n = 3N$

$\Gamma(n) = n!$ for integer n

$$\Omega(E) \propto (2mE)^{\frac{3N}{2}} V^N$$

↑ proportionality constant only depends on N

technical note: $\Omega(E)$ should be dimensionless

but it has dimensions (pos x momentum)^{3N}

a proper quantum treatment would quantize phase space

in units of phase space area h

$$\dots \propto \left(\frac{3N}{2} \right)^{1/2} V^N$$

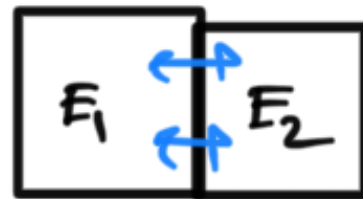
so $\Omega(E) = \frac{(2\pi E)^{3N}}{h^{3N} V}$ dimensions

$$S(E) = \frac{3N}{2} \ln E + N \ln V + \text{const}$$

Entropy of an ideal gas
of energy E and volume V

Step 2 statistical mechanics of two interacting systems

- let's say you have systems 1 and 2 with energy E_1 and E_2
- you bring them into thermal contact so they can exchange energy



- but combined system is closed $\Rightarrow E_1 + E_2 = E_T$ $E_T = \text{total energy is constant}$
- by exchanging energy what will E_1 and E_2 settle down to at equilibrium?

$$\Omega_T(E_1, E_2) = \Omega_1(E_1) \Omega_2(E_2)$$

$$S_T(E_1, E_2) = S_1(E_1) + S_2(E_2)$$

$$E_2 = E_T - E_1$$

2nd law of thermodynamics

- E_1 and E_2 will change so as to maximize total entropy
or # of states explored : $\Delta S_T > 0$
- equilibrium occurs when S_T is maximum

Maximum entropy
principle for
closed systems

$$\frac{\partial S_T}{\partial E_1} = \frac{\partial S_1}{\partial E_1} + \frac{\partial S_2}{\partial E_2} \underbrace{\frac{\partial E_2}{\partial E_1}}_{-1} = 0$$

$$\frac{\partial S_1}{\partial E_1} = \frac{\partial S_2}{\partial E_2}$$

condition for equilibrium

$$\text{let } \frac{1}{kT} = \frac{\partial S}{\partial E}$$

definition of temperature T

kT has units of energy

$$T_1 = T_2$$

Condition for equilibrium is equality of temperature :-

... definition of temperature make intuitive sense? check for ideal gas:

recall: $S(E) = \frac{3N}{2} \ln E + N \ln V + \text{const}$

$$\frac{1}{kT} = \frac{\partial S}{\partial E} = \frac{3N}{2} \frac{1}{E} \Rightarrow \boxed{E = \frac{3}{2} N kT}$$

energy is proportional to temperature ✓

interpretation:

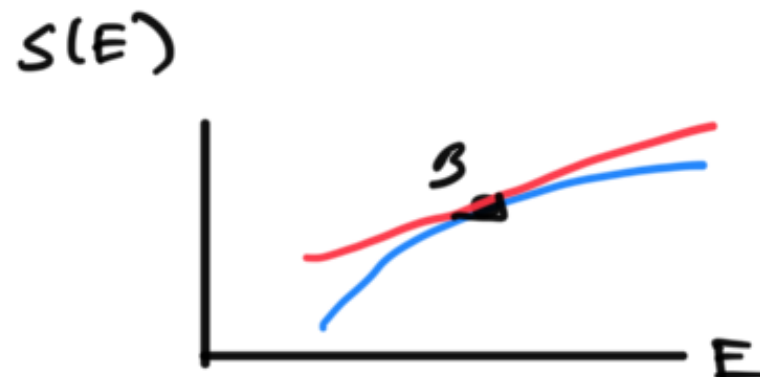
- each particle moves in 3 dimensions
- so has 3 degrees of freedom
- there are N particles

- so $\boxed{\text{Energy per degree of freedom is } \frac{1}{2} kT}$

this is a more general statement:

energy per each degree of freedom
that has a quadratic energy is $\frac{1}{2} kT$

geometrically:

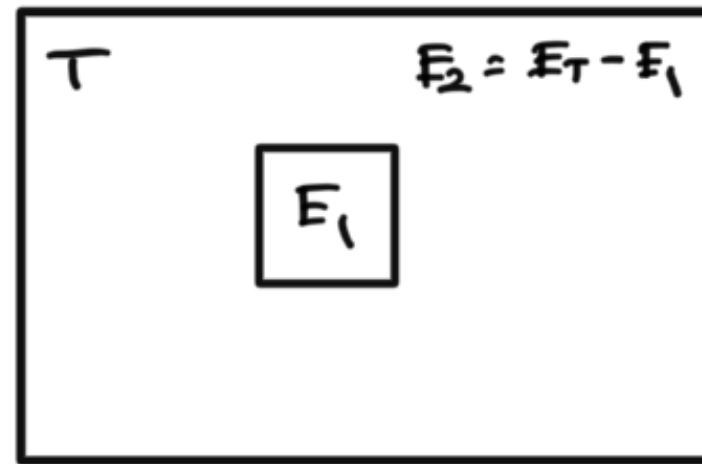


$$\beta \equiv \frac{1}{kT} = \text{slope of } S(E)$$

- $S(E)$ concave
- $\Rightarrow$ slope β dec with E
 - $\Rightarrow$ temp kT inc with E

Step 3 understand Statistical Mechanics of open systems

- now let system 1 be a system we care about
system 2 be a large reservoir it is in equilibrium with at temp T



- system 1 can exchange energy with the reservoir
- what is the probability system 1 is in microstate $\vec{s}$ with energy $E_1(\vec{s})$

$$p(\vec{s}) \propto \underbrace{\Omega_2(E_T - E_1(\vec{s}))}_{\text{def of entropy}} = e^{S_2(E_T - E_1(\vec{s}))}$$

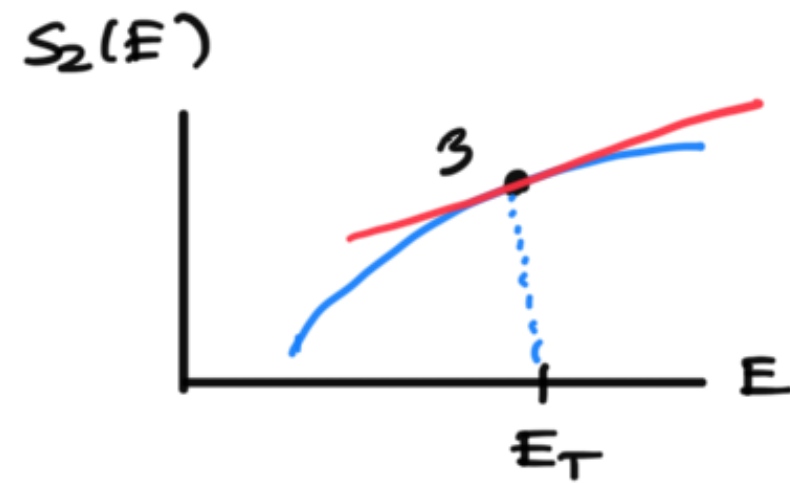
$\underbrace{p(\vec{s})}_{\text{prob sys 1 is in microstate } \vec{s}} \propto \underbrace{\Omega_2(E_T - E_1(\vec{s}))}_{\text{\# states reservoir can explore}}$

- now assume $E_1 \ll E_T$ (reservoir is large and system 1 is small)

$$\beta = \frac{1}{E_T} \left[\frac{\partial S_2}{\partial E} \right]$$

• Taylor expand

$$S_2(E_T - E_1) \approx S_2(E_T) - \left(\frac{\partial S_2}{\partial E} \right)_{E_T} E_1$$



• so $p(\vec{s}) \propto e^{S_2(E_T) - E_1(\vec{s})/kT}$

$$p(\vec{s}) \propto e^{-E_1(\vec{s})/kT}$$

intuition

- $\vec{s}$ has higher energy
- $\Rightarrow$ reservoir has lower energy
- $\Rightarrow$ reservoir explores exponentially fewer states
- $\Rightarrow p(\vec{s})$ exponentially suppressed

put in normalization: $\int d\vec{s} p(\vec{s}) = 1$

$$p(\vec{s}) = \frac{1}{Z} e^{-E_1(\vec{s})/kT}$$

$$Z = \int d\vec{s} e^{-E_1(\vec{s})/kT}$$

Boltzmann distribution over the microstates $\vec{s}$ of a system in equilibrium with a reservoir of temperature T